

AN OCEAN OF GRADIENTS — TOWARDS A 3D MAPPING AND VISUALIZATION OF HIGH-RESOLUTION MARINE DATA

AN OVERVIEW ⁽¹⁾

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1. BACKGROUND

Understanding the spatial heterogeneity of marine environments is fundamental to comprehending the complex dynamics of ocean ecosystems. The ocean is characterized by pervasive variations in key properties such as temperature, salinity, and nutrient concentrations, forming what can be conceptualized as an “ocean of gradients”. These gradients play a crucial role in structuring habitats and influencing the distribution of marine life (Nezlin 2010). Effectively analysing and interpreting the increasing volume of high-resolution marine data necessitates the use of sophisticated tools capable of visualizing and quantifying these multi-dimensional gradients.

Geographic Information Systems (GIS) and related software have become indispensable for exploring the spatial complexity of ocean gradients. These tools allow researchers to integrate diverse datasets, including bathymetry, surficial geology, and hydrographic information, to create comprehensive representations of the marine environment (Kostylev *et al.* 2001). By visualizing data in a spatial context, scientists can identify critical habitat features and understand how environmental gradients influence their characteristics (Tufekci *et al.* 2010). For instance, mapping benthic habitats often relies on overlaying bathymetric data with information on sediment type and water column properties, revealing how gradients in these factors contribute to habitat complexity and suitability for different species.

Furthermore, the analysis of oceanographic gradients extends beyond static spatial representations. Understanding the interplay between different environmental variables and their combined influence on marine ecosystems requires tools that can handle multi-layered data. By integrating data on temperature, salinity, nutrients, and other relevant parameters, researchers can gain insights into the complex relationships that drive ecological patterns (Nezlin 2010). The ability to visualize these gradients in a spatially explicit manner allows for the identification of key ecological boundaries, such as oceanic fronts and upwelling zones, where sharp changes in environmental conditions can lead to distinct biological communities.

In conclusion, the study of ocean gradients is crucial for advancing our understanding of marine ecosystems. The development and application of sophisticated tools, including GIS and specialized oceanographic software, enable researchers to effectively analyse and visualize the multi-dimensional nature of these gradients. By integrating diverse datasets and employing spatial analysis techniques, scientists can unravel the complex relationships between environmental variability and the distribution of marine life, ultimately contributing to more informed conservation and management efforts (Kostylev *et al.* 2001; Tufekci *et al.* 2010; Nezlin 2010).

2. DEFINING OCEAN GRADIENTS

Having discussed the widespread presence of oceanic gradients and the utility of tools such as GIS in their analysis, it's vital to precisely define what a “gradient” entails. This section, therefore, broadens the concept beyond basic interpretations, exploring its multifaceted spatial, temporal, and variable characteristics.

The Merriam–Webster dictionary defines “gradient” as “the change in the value of a quantity (such as temperature, pressure, or concentration) with change in a given variable and especially per unit distance in a specified direction” (Merriam–Webster n.d.). While this definition offers a foundational understanding, it necessitates further elaboration to enhance its clarity and applicability across diverse contexts.

The term “distance” in the definition is particularly crucial and warrants specific attention, as it can encompass various forms of measurement. It may refer to Euclidean distance or another metric. Importantly, “distance” can also denote temporal separation, such as observations taken at different times, or represent a spatiotemporal dimension. This is evident in phenomena like the El Niño-Southern Oscillation (ENSO), a large-scale recurring ocean-atmospheric phenomenon where observations are recorded across various locations and times. The variability in distance scales, from micro to macro, ultimately depends on the specific research question.

Moreover, the phrase “change in a given variable” also requires expansion. This variable can be any directly observable quantity (Figures 1a, 1b), such as temperature readings from a conductivity-temperature-depth (CTD) profile—a tool that records vertical structure of marine systems. Alternatively, the variable might be an inferred measure derived from primary observations, like the variance of a measured quantity (Figure 1c, 1d) or fluctuations in periodic components of a system (Figure 1e), as observed in Quaternary sea-level curves around 800 kyr ago (Paillard 2001).

Beyond individual measurements, entire systems or subsystems within a larger framework can undergo either abrupt shifts or gradual transitions from one state to another over time. These

changes are often termed “tipping points,” representing critical moments where a system’s behavior abruptly shifts, leading to significant alterations in its dynamics. Tipping points can be viewed as manifestations of gradients, as they reflect pronounced changes in the system’s state variables across spatial or temporal scales. These shifts may arise from nonlinear feedback mechanisms that can amplify small perturbations, pushing the system towards a new stable state or an unstable condition. Changes in feedback amplitude, periodicity or directionality constitute a type of gradient that is beyond directly measured variables and it is well suited to describe the complex relationship present at the ecosystem and planetary scale. Understanding tipping points and their relationship to gradients is essential for predicting and managing the complex behavior of natural systems, particularly under anthropogenic forcing.

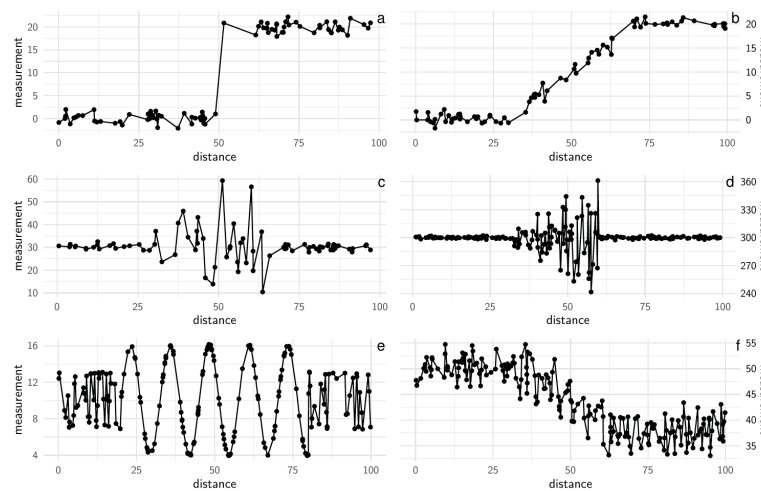


Figure 1. These six scenarios illustrate various simulated gradients: (a) a distinct shift in the overall level; (b) a period of stability, followed by a linear increase, and then a return to a new stable level; (c) and (d) a noticeable change in data variance occurring after a critical point, with the variance subsequently returning to its original state. This example highlights the concept of “temporal distance.” (e) changes in the periodic component of a variable; (f) variability is stable over time but the system is slowly drifting toward a new state. The figures are inspired by Lindeløv (2020), and the corresponding code for reproduction can be found at https://github.com/zboraon/gradients_monograph_simulation/tree/main.

In refining these concepts, it becomes apparent that the definition of a gradient extends beyond merely measuring changes across distances in physical space. It also involves changes across temporal and spatiotemporal scales and can encompass both explicit and implicit changes in observed variables. This broadened perspective is crucial for accurately describing gradients in a scientific context.

The concept of a gradient applies broadly to geology and geological features, encompassing both their manifestation and their causes. A prime example is the variation in temperature with Earth’s depth, known as a geothermal gradient or geotherm. Geothermal gradients can change spatially due to the intrinsic properties of the rock, such as the concentration of heat-producing radiogenic elements. They can also shift in response to localized events like tectonic activity. Beyond these directly measured variables, dynamic changes in a system’s feedback loops—such as their amplitude, frequency, or direction—also constitute a type of gradient. This concept is well-suited for describing complex relationships at both the ecosystem and planetary scales,

particularly in relation to the Earth's internal systems (*e.g.*, magma and geothermal processes).

Nevertheless, the concept of gradient extends to a much wider variety of physicochemical and spatiotemporal features. This includes, but is not limited to, pressure variations, rock and fluid compositions, mechanical properties, tectonic stresses and strains, porosity and permeability, the state of and distance from thermodynamic equilibrium (*e.g.*, metastable, unstable, stable), and geomorphological features. Additionally, the geological concept of a gradient is intrinsically scale-dependent, with detectable gradients ranging from the nanoscale to the scale of the Earth itself. Geological gradients are also inherently transient or evolving, with rates and lifetimes spanning from geologically instantaneous phenomena, such as strain gradients during a fault-localized seismic event, to millions or billions of years, like lithospheric-scale, plate tectonic-driven modifications.

Geological manifestations, such as rocks, magmas, sediments, faults, and gravitational processes, can be either the cause or the consequence of gradients. For instance, chemical gradients in rocks and/or fluids can arise from initial compositional disequilibria between two neighboring materials or result from the re-equilibration of those materials over space and time. The interconnections among this vast range of parameters, scales, and times can determine or be the consequence of geological gradients at all spatial and temporal scales. Local phenomena, whether in space or time, can profoundly alter larger and longer-lived processes and their associated gradients, imposing feedbacks upon them.

Gravity is a primary driver of Earth gradients and controls several processes contributing to the differentiation and stratification of the crust, ocean and atmosphere. In complex systems, the effect of gravity allows for the creation of complex dynamic (both spatially and temporally) gradients controlling complex geo-biological feedback. Winogradsky columns are a great example of the power of gravity in structuring layers. Winogradsky columns are microbial microcosms conceived by Sergei Winogradsky in the 1880s to reproduce complex microbial ecosystems in the laboratory. They are a great example of the power of gravity in structuring layers. The column's construction starts with preparing a homogenous mixture of natural samples (typically soil or sediments) and water, which is either collected from the natural site or designed in the lab to enrich specific microbial metabolisms. The mixture is then poured into a high aspect-ratio flask, closed at the top and exposed to light. Gravity will immediately start to act on the system favoring size sorting within the sediment column, compacting the sediment grains and favoring the ascent of the gases produced in the deeper layers by anaerobic microorganisms. Gravity and light will favor the presence of specific microniches controlled by geochemical gradients. Within a few days to weeks the column will develop strong geochemical and biological gradients visible as distinct colored layers within the column. The geo-biological spatial gradient developed will change in time providing temporal gradients (Figure 2).



Figure 2. Picture of two months old Winogradsky columns obtained from sediment of the coastal brackish Lucrino Lake near Naples (Italy). The distinct color represents diverse microbial communities occupying distinct ecological niches created by the interaction of gravity, light and microbial metabolisms. The spatial geo-biological gradients are everchanging and provide temporal gradients as well. The different columns started with the same sediments and water, but were amended with different nutrients. Columns realized in the Giovannelli Lab at the University of Naples Federico II.

Biology can directly influence both physical and chemical environmental parameters. Life, especially microbial life, is capable of interacting with a large swath of the periodic table and can oxidize and reduce a very large number of molecules. The accumulation, diffusion and transport of these molecules can create strong local gradients that in turn influence other organisms. This happens for example in the sediments hosting *Posidonia* prairies where sulfate reducing bacteria use the high organic carbon accumulating in the rhizome and produce hydrogen sulfide creating strong sedimentary gradients influencing the distribution of all other burrowing organisms. Similarly, the ability of microbial life to precipitate and dissolve a large number of minerals can alter the physical properties and the flow patterns of fluids in sediments and rocks in the subsurface hosting complex microbial communities. A last example is constituted by the ability of microbial life to create and control the distribution of hyperacidic environments. Environments with pH below 3 on our planet are widespread and generally linked to the presence of sulfide minerals or degassing magmatic chambers. In the presence of sulfide minerals exposed to the surface, like in areas of intense mining activities, specialized microbial consortia oxidize sulfide minerals to sulfuric acid therefore acidifying the surrounding pH to levels toxic to most other life. There are just a few examples of the complex interactions through which biology and geology create and control environmental gradients.

In the Mediterranean Sea, several decades of both sparse and collected geophysical data—including bathymetric, gravimetric, seismic, and magnetic information—reveal a profound diversity of seafloor geological features, horizontally and vertically. Some data, like heat flow data, are available across the entire Mediterranean surface at rather constant resolutions.

Others, such as lithological features, are sparser and tied to localized direct investigations, such as a few boreholes. Combining and visualizing these existing data, whether sparse or continuous, and both onshore and offshore, provides an invaluable tool to predict and guide investigations into the geology of underwater Mediterranean features, their associated gradients, and their potential control over biological, chemical, and physical manifestations. Furthermore, biological processes and evolutionary adaptations significantly shape and respond to these oceanic gradients, from microbial community distributions influenced by nutrient availability to the emergence of distinct faunal zones driven by pressure and light variations.

3. PRACTICAL WAYS TO USE AVAILABLE RESOURCES

Understanding the ocean's intricate, ever-changing nature requires advanced computer tools. This section explores innovative ways to use technology for better 3D mapping and visualizing marine data. This includes grouping similar environmental areas (what we call “environmental niches”), simplifying vast ocean genetic data (metabolomics pathways), and developing a Microbiome MetaCube (MMC) prototype. These new tools help us bring together various types of data into one easy-to-access system.

3.1. Accessing data

Getting access to diverse marine data is key to understanding and visualizing ocean gradients—those changes in conditions over space or time. We do this by using well-known global data portals like GEMCO, EMODNET, MGnify, EarthChem, and ODATIS, as well as project specific data repositories, such as TARA and MALASPINA, which offer a wide range of large, publicly available datasets. We also use data from more specialized, research-focused initiatives, which often provide specific datasets or have their own dedicated portals. Good examples include data from Deep Hypersaline Anoxic Basins (DHABs), Hydrothermal Vents, the CoEvolve project and the eGEOTRACES portal (which is part of the larger GEOTRACES program and helps visualize its datasets). All these different data sources are excellent for describing and showing these ocean gradients.

Beyond these broad portals, effective data access also means making specific searches within large biological and geochemical databases, such as those holding metagenome data (genetic material from entire communities of microbes). This lets us pull out targeted, interdisciplinary information on various elements vital to marine systems. Such information components may include concentration of redox-active elements, metals, carbon, different pollutants (like plastics), and even pathogens. These combined data access strategies are essential for creating advanced computer tools for managing marine environments, which we'll explain in detail in the next subsections.

3.2. IT tools

3.2.1. Clustering environmental niches

Each geographic location where data is collected acts as a unique example, helping to define the specific characteristics of the living organisms found there—this is their biological niche. By gathering and then grouping these data points, the objective is to build a comprehensive database of Niche Profiles (NP). The idea is to put identical ocean niches into the same NP

group. This fundamental concept isn't just for the ocean; it applies to any ecological setting.

A tool under development is designed to process physical and chemical data obtained from public repositories, with every data point carefully described. Once this complete dataset is assembled, rigorous mathematical modeling is then applied to create an NP classifier—essentially, a system that categorizes these niches.

A crucial part of such a tool's development is its ability to work seamlessly with other oceanographic data visualization software like Ocean Data View (ODV) and the pilot CIESM 3D visualization tool (as described by Rassoulzadegan in this volume). The physical and chemical data drawn from public repositories—used for defining biological niches and building the NP classifier—can be easily connected with ODV and the pilot CIESM 3D visualization tool, among others. This interoperability (the ability for different systems to work together) lets users directly import the generated niche profiles and related data for advanced visual exploration, analysis, and mapping.

This type of tool and its database are expected to produce a significant side benefit: a detailed inventory of marine biological niches. This inventory can potentially be linked with the information presented in Section 3.2.3 (Microbiome MetaCube).

3.2.2. Simplifying ocean microbial metabolic pathways

Metagenomic data (genetic material from entire microbial communities) from the Mediterranean Sea can be analyzed using MetaDAG. This is a new bioinformatics tool specifically designed to model and simplify the complex metabolic networks—essentially, all the chemical reactions—that occur within organisms and whole microbial communities (biomes). Other specialized tools can be used to annotate large metagenomic collections to obtain information on the taxonomic and functional diversity of each sample, such as DRAM, Metabolic or Geomosaic.

In order to uncover which types of microbes are present and what they're doing, specialized tools like Kaiju, SqueezeMeta, PiCrust, GTDBtk, or q2-metnet can be employed. Functional annotation of the metagenomes can be accomplished using a large number of tools, including funprofiler, DRAM, KOfamScan, KEGG, or database specific searches using Diamond or HMM. Several of these tools are packaged into pipelines or available through the Galaxy or KBASE platform. The choice of tool depends on how the genetic information was originally sequenced and the specific objectives of the research. Several of these tools help identify specific KEGG Orthology (KO) gene IDs within the samples, which are like unique barcodes for certain genes involved in metabolic processes. Based on these KO IDs, the metabolic networks can be reconstructed into detailed diagrams called reaction graphs, then into directed acyclic graphs (DAGs), and finally into simplified versions known as compressed metabolic directed acyclic graphs (metaDAGs).

Another important use of this method is building the core and pan-metabolism of these microbial communities. The “core metabolism” refers to the essential chemical reactions shared by most microbes in a sample, while “pan-metabolism” includes all the reactions present across all samples combined. Once data from all relevant samples are collected, the pan-metabolism of Mediterranean microbiomes can be assembled. This comprehensive metabolic dataset is currently being developed and will form the foundation for building a MetaCube.

3.2.3. *Microbiome MetaCube (MMC)*

A universal way to represent metabolic pathways, those intricate chains of chemical reactions that sustain life, is required. While current tools rely on metabolic maps and network arrangements, challenges persist in comparing and combining these connections, especially in three-dimensional environments where ocean conditions, for example, change continuously, creating diverse gradients.

Some researchers (like Reyes-Prieto *et al.*, discussed in this volume) are actively developing a tool to create standard 3D maps of metabolic pathways. These maps, when layered with metagenomic data from global genetic databases, are designed to highlight the specific metabolic abilities linked to different genetic data points. These 3D representations of metagenomic data could also aim to bring together geological, physical, chemical, and biological data into a single, globally accessible system. This helps contribute to wider efforts in digital ocean mapping and modeling. For example, core and pan-metabolic data described in Section 3.2.2 (Ocean's Metabolomics Pathways Compression Tool) can be visualized within the MMC (MetaCube) environment, as described in Section 5.2.

3.2.4. *Integrating physical, geochemical and metabolic data*

An urgent need exists to develop innovative visualization and analysis tools that go beyond those currently available and those described in this volume, integrating data points from the physical, chemical, geochemical, and biological domains in novel ways. These tools must address the complexity of natural ecosystems in a truly interdisciplinary fashion. Their primary purpose will be to aid the interpretation of complex geobiological feedbacks that characterize the ocean and to carry predictive power useful for understanding the ecosystem's response to future changes, thereby defining new theoretical and practical approaches to investigation.

4. SAMPLING AND DETECTING GRADIENTS OF RELEVANCE FOR MICROBIAL LIFE

A deeper understanding of microbial life requires moving beyond measures of total biomass to focus on the fundamental units of organization: microbial communities and their interactions. These communities, rather than sheer abundance, drive the net process rates underlying biogeochemical cycles that sustain life on Earth. Capturing the spatial and temporal dynamics of microbial community structure—alongside the gene expression patterns that mediate trophic interactions—is therefore essential to a comprehensive understanding of ecosystem function. The spatial organization of microorganisms, shaped by environmental gradients and evolutionary histories, remains largely unmapped. Uncovering these patterns will require intensive, systematic surveys of microbial genetic and functional diversity across a wide range of ecological and geological contexts.

Here follow two examples of standardized, integrated analyses of the tremendous complexity of natural microbial systems based on innovative technology and a shared vision: to fully understand genomes, they must be studied within the environmental frameworks in which they evolved, function, and continue to adapt.

4.1. Sampling extreme marine interfaces: a methodological imperative for microbial research

Understanding the distribution and functional roles of microbial communities in extreme marine environments demands precise, high-resolution sampling. This is especially true along the steep physicochemical gradients that define the interfaces between seawater and hypersaline brines. These interfaces, often less than two meters thick, are ecological hotspots where intense chemical stratification supports unique microbial metabolisms, like chemolithoautotrophy. However, accessing and accurately sampling these gradients at depths exceeding 3,000 meters poses significant technical challenges.

The need for accurate, gradient-resolving sampling is perfectly exemplified by studies on Deep Hypersaline Anoxic Lakes (DHALs) in the Eastern Mediterranean Sea. Formed from the dissolution of ancient Messinian evaporites, these brine lakes are isolated, high-salinity ecosystems confined within deep depressions. Their exceptional chemical variability and sharp density stratification create extreme, layered environments, ideal for studying microbial adaptation and community structure.

To target these unique environments, exploration begins with high-resolution digital terrain models (with a resolution of 500 meters or less) to pinpoint confined depressions likely to host brines. These candidate basins are then further investigated using 3.5 kHz Chirp swath-bathymetry profilers. These instruments detect the brine-seawater interface by capturing sharp acoustic reflection lines resulting from the high salinity contrast, typically sealed by the basin walls.

Once a promising site is identified, in situ verification is performed using a rosette sampler equipped with conductivity-temperature-depth (CTD) sensors. Key physical parameters—salinity (via conductivity), temperature, and oxygen—are recorded and depth-calibrated to precisely locate the interface and determine its thickness. Sampling at this stage remains qualitative; the exact chemical composition must be confirmed through subsequent laboratory analyses.

Only after this multi-step verification can the interface be effectively sampled. In addition to the traditional single-point sampling approach, a specialized method allows for 10 cm-resolution sampling across the interface layer. This technique ensures discrete collection along the steep salinity gradient.

The process involves several meticulous steps outlined in Figure 3:

1. Initial Descent and Stabilization: The rosette, carrying Niskin bottles and CTD sensors, is slowly lowered (at 0.3—0.6 m/s). The CTD sensor precisely indicates the interface's vertical position by a significant increase in conductivity. The rosette continues into the brine for another 25 meters, then pauses to straighten the cable and stabilize its vertical position.
2. Precise Withdrawal and Bottle Closure: The rosette is then slowly withdrawn from the brine at a speed of less than 0.1 meters per second. Niskin bottles are triggered to close at the identified interface position, ensuring samples are collected from specific, narrow bands within the gradient.

This method provides intact stratified samples, which are crucial for downstream microbiological, geochemical, and genomic analyses.

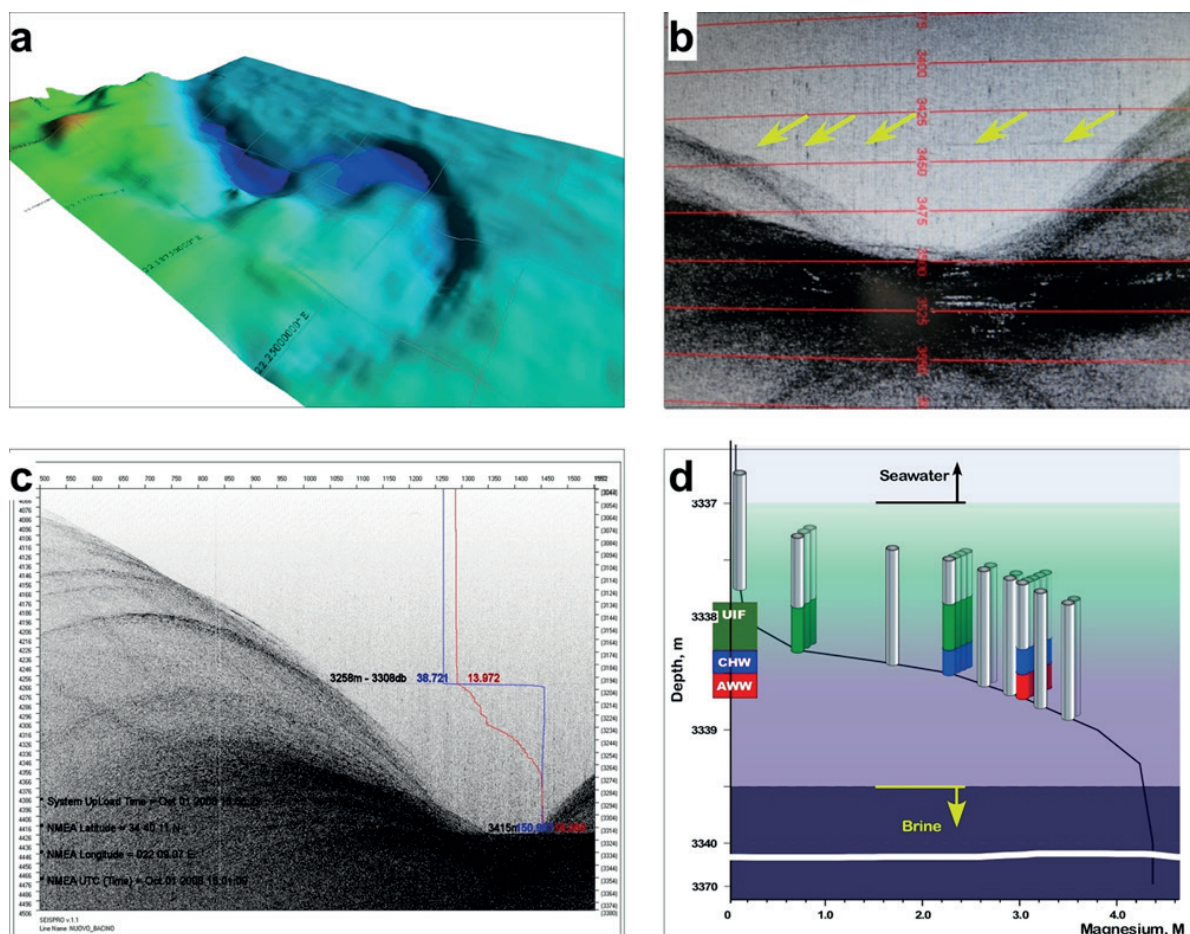
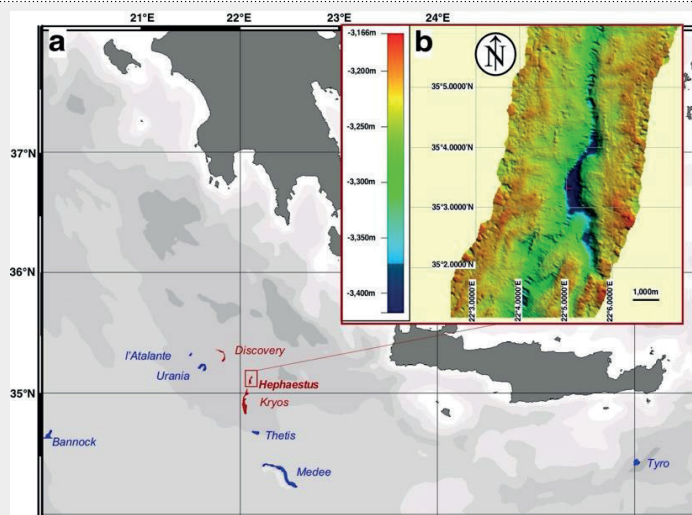


Figure 3. Four key steps in exploring the sharp physicochemical gradients at the seawater-brine interface of Mediterranean Deep Hypersaline Anoxic Lakes (DHALs): (a) Identifying deep, confined depressions using a morpho-bathymetric digital terrain model of the Mediterranean Sea to pinpoint potential DHAL locations. (b) Utilizing 3.5 kHz Chirp swath-bathymetry profiling with multibeam mapping to reveal a perfectly straight reflection line (yellow arrows), indicating the interface. (c) Performing an initial cast with a CTD-equipped rosette to measure essential parameters (conductivity/salinity and oxygen concentration) that highlight the sharp physicochemical gradients. (d) Implementing an interface subsampling strategy with Niskin bottles mounted on a CTD-rosette.

This refined protocol was successfully applied in the identification and characterization of the Hephaestus brine lake, one of the most saline natural hydrological formations known. Detected approximately 10 km north-northeast of DHAL Kryos, the lake is hosted within a steep, sinuous fracture reaching a depth of 3,423 meters. The presence of a sharp brine-seawater interface was first indicated by bathymetric and acoustic data, and later definitively confirmed through CTD profiling and targeted brine sampling. This case vividly illustrates the critical need for high-precision, adaptive sampling methodologies to explore the unique microbial ecosystems associated with marine brine lakes. Without such rigor, the steep physicochemical gradients that define these habitats—and the microbial communities they support—would remain inaccessible, under-characterized, and disconnected from broader biogeochemical understanding.

Deep-sea, hypersaline, anoxic lakes (DHALs)

in the Eastern Mediterranean, likely formed through main geological processes associated to tectonic events, which led to the submarine dissolution of exposed 0.9 to 5.3-million-year-old Messinian evaporites. The chemistry of the DHAL brines is highly variable, primarily due to the dissolution of different layers within the Messinian evaporitic suite (La cono *et al.* 2011; Yakimov *et al.* 2014).



Most Mediterranean DHALs are thalassohaline, meaning they have a seawater-like proportion of major ions, but with a total salinity 7–10 times higher than that of normal seawater. Due to the density difference of about 20–30% between brines and seawater, DHALs are separated from the overlying seawater by a sharp interface, typically less than 2 meters thick, characterized by very steep chemical gradients and considered a hotspot for biological activity, including chemolithoautotrophy.

4.2. Metagenomics at the crossroads of medical and environmental research

Understanding the intricate interplay between microorganisms and their physicochemical environments—whether in natural ecosystems or within the human body—has become a central focus in microbiome research. Thanks to advancements in experimental methodologies and analytical tools, we can now precisely characterize microbial communities and the environmental gradients that shape them.

For example, metataxonomy (also known as metabarcoding), which uses taxonomic marker genes, and metagenomics (or shotgun sequencing), which captures a sample's entire genomic content, are rapidly transforming our understanding. These approaches place microbes within the environmental frameworks in which they evolved, function, and continue to adapt. Metagenomic approaches, in particular, capture the complexity of microbial communities in situ, offering insights into the spatial and temporal dimensions of microbial diversity, along with the gene expression patterns driving trophic interactions and system-level functionality.

Crucially, both approaches are applicable across environments, enabling a comparative framework that spans oceanographic samples and human-derived material. The basic workflows for DNA extraction are similar regardless of sample origin, though protocols

adapt for differences in biomass density, matrix composition, and preservation needs. Table 1 summarizes key considerations in DNA extraction techniques across different settings.

In human microbiome studies, stool samples are commonly used due to their ease of collection, high microbial load (~ 10^6 cells/mL), and straightforward preservation—typically by freezing at -20°C . RNA-focused studies often incorporate stabilizing agents prior to extraction. For environmental samples, where microbial biomass may be more diluted or spatially heterogeneous, field-deployable filtration systems and preservation protocols have been developed to capture and stabilize the microbial signal across diverse physicochemical gradients.

Step	Human Gut Microbiome	Free-Living Environmental Niches
1. Sample Collection	Collect stool samples, swabs, or biopsies from the gut or other matrices.	Collect soil, water, air, or surface samples.
2. Dilution/Concentration		Dilute or concentrate samples to achieve optimal microbial concentration, avoiding/reducing chemical contaminants which may impair the PCR and library preparation steps.
3. DNA Extraction	Extract DNA from the collected samples using usually specialized kits.	Extract DNA from environmental samples using ad-hoc adapted methods or specialized kits.
4. Primer Selection	Choose appropriate primers targeting specific regions of the 16S rRNA gene or other relevant genes to ensure comprehensive coverage and minimize biases.	Choose appropriate primers targeting specific regions of the 16S rRNA gene or other relevant genes to ensure comprehensive coverage and minimize biases.
5. Library Preparation	Prepare DNA libraries for sequencing, including amplification and barcoding.	Prepare DNA libraries for sequencing, including amplification and barcoding.
6. Sequencing	Perform sequencing using platforms like Illumina or PacBio.	Perform sequencing using platforms like Illumina or PacBio.
7. Data Analysis & Visualization	Analyze sequencing data to identify microbial taxa and functions. Visualize results using bioinformatics tools.	Analyze sequencing data to identify microbial taxa and functions. Visualize results using bioinformatics tools.

Table 1. Considerations for microbiome analysis comparing methodological steps for human gut and free-living environmental samples.

These studies demonstrate that the human microbiota, much like microbial communities in oceans, soils, or sediments, plays a critical role in regulating key metabolic processes. In both cases, microbial communities aren't passive residents but active agents in maintaining

system homeostasis—be it ecosystem function or human health. Disruptions in these microbial systems, whether due to pollutants in nature or antibiotics and dietary shifts in humans, can trigger dysbiosis, undermining the resilience and functionality of the host environment.

New techniques such as single-cell genomics, spatial transcriptomics, and in situ hybridization are extending the analytical frontier. These allow researchers to examine microbial function and identity at the level of individual cells and in direct relation to their micro-environmental conditions.

Finally, the success of molecular analyses heavily depends on the computational tools used to process and interpret sequencing data. Metataxonomy remains widely adopted due to its cost-effectiveness and the relative maturity of its bioinformatics pipelines. However, its performance is influenced by primer design, which must be carefully chosen to ensure broad and unbiased taxonomic coverage for the target sample.

Overall, the integration of advanced sampling strategies, high-throughput molecular techniques, and powerful bioinformatics platforms is opening new pathways for characterizing microbial communities and the gradients that structure them—whether in the open ocean, coastal sediments, or the human gastrointestinal tract. This convergence of technology and ecological insight is central to the next generation of microbiome science.

5. PROTOTYPE EXERCISES TO VISUALIZE GRADIENTS IN 3D

The pilot 3D visualization tool being developed under the auspices of CIESM (Rassoulzadegan, this volume) represents a significant advancement in marine data analysis. This versatile and user-friendly tool was developed to improve upon previous limitations in assessing gradients within the Mediterranean Sea. It facilitates the visualization and interpretation of complex marine data, including biological, physical, chemical, and even genetic parameters. By allowing for the overlay of these diverse datasets onto 3D bathymetric maps, among other features, the tool enables volumetric visualization of nutrient hotspots and enhances the detection of biogeochemical gradients. This capability is crucial for understanding the intricate interactions and environmental changes occurring across the Mediterranean Sea. Below, two examples will be presented to demonstrate how CIESM scientists collaborate to test this tool through prototypes, visualizing and analyzing physicochemical, nutrient, and genetic data to gain a more comprehensive understanding of the marine environment.

5.1. Physicochemical parameters and nutrients

The pilot CIESM 3D visualization tool is powerful for displaying interactions, geochemical gradients, and areas of high reactivity and elemental flux within marine systems. This approach significantly enhances our ability to assess gradients intuitively compared to other methods. Specifically, one of its prototypes allows for the volumetric visualization of nutrient hotspots and improves the detection of biogeochemical gradients by overlaying physicochemical and nutrient data onto 3D bathymetric maps. These visualizations are most useful for identifying both vertical and horizontal environmental gradients across the Mediterranean Sea. The distinct features of the Mediterranean basins, along with Atlantic water inflow, justify using both 2D and the pilot CIESM 3D visualization tool's mapping techniques to effectively display these gradients.

Biogeochemical processes, involving the transformation of organic and inorganic components, are influenced by factors like temperature, salinity, photosynthetically available radiation (PAR), and other biogeochemical properties, all contributing to environmental gradients. Temperature is a key driver in assessing these gradients. The CIESM 3D visualization presented in this section clearly shows the Mediterranean's temperature gradient: cooler waters (16–18°C) in the west and warmer waters (21–22°C) in the east, particularly in the Levantine Basin and near Egypt and southern Turkey (Figure 4). This visualization helps interpret how oceanographic factors, such as Atlantic water inflow and regional circulation, shape this gradient.

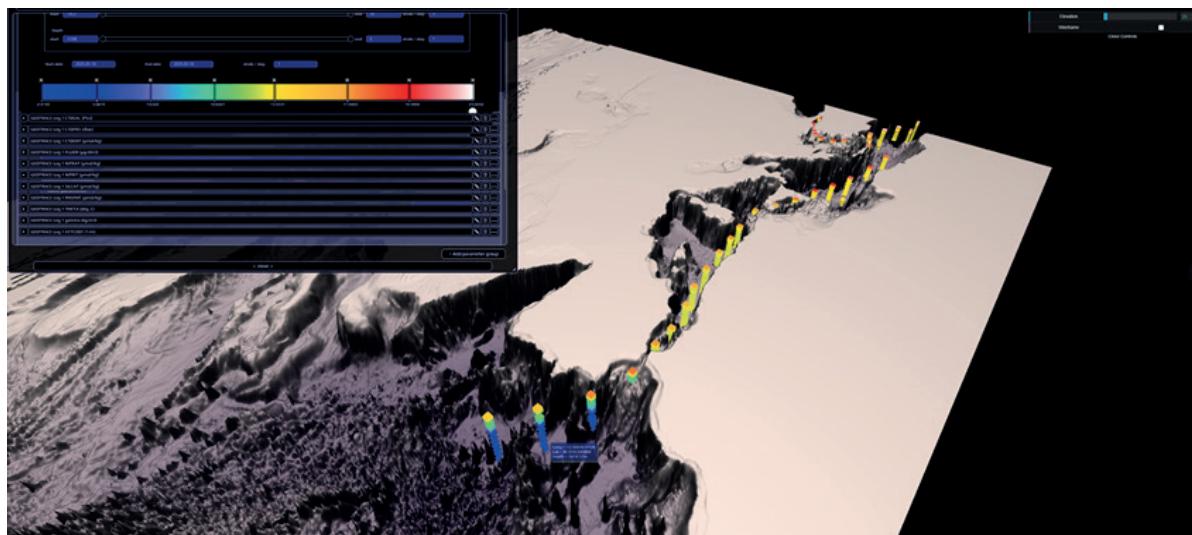


Figure 4. Temperature (C°) in the Mediterranean Sea along the Southern Leg (S) of the GEOTRACES-A04N section during R/V Pelagia cruises (64PE370/374) in 2013.

The CIESM 3D visualization significantly aids in understanding and interpreting nutrient distributions. For instance, while salinity plots show distinct water masses and nutrient distributions indicate oligotrophic conditions in the eastern basin, the CIESM 3D visualization of nitrate concentrations in the 0–50 m layer (Figure 5) aligns well with modeling results. This tool facilitates the interpretation of a clear gradient in deeper waters, with higher nitrate concentrations near the Atlantic inflow at the Strait of Gibraltar, gradually decreasing eastward.

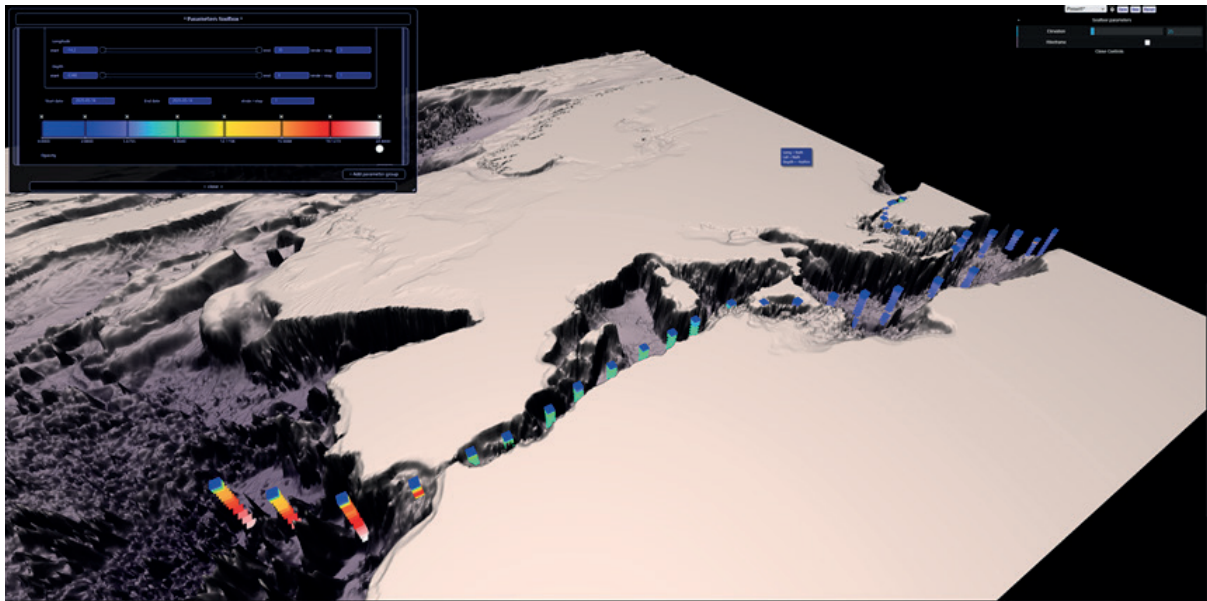


Figure 5. Nitrates ($\mu\text{mol/Kg}$) in the Mediterranean Sea along the Southern Leg (S) of the GEOTRACES-A04N section during R/V Pelagia cruises (64PE370/374) in 2013.

This highlights how the pilot CIESM 3D visualization tool doesn't just display data; it also helps us understand complex nutrient patterns, like the decrease in concentrations along the anti-estuarine circulation from the Strait of Gibraltar, where nutrient-poor Atlantic Water enters. We can also effectively visualize and interpret the main sources of dissolved silicate in the Mediterranean Sea, including continental river inputs and submarine groundwater discharge, using these 3D models.

5.2. Human gut metabolomics

The pilot CIESM 3D visualization tool proves to be an exceptionally powerful resource for exploring the intricate details of complex biological systems, including microbial metabolic networks within the gut. This advanced tool significantly enhances our ability to analyze and interpret the vast datasets generated in microbiome research, moving beyond traditional 2D representations to offer a truly immersive and insightful view of microbial functions. Building on this foundation, the MetaCube application, a prototype of the pilot CIESM 3D visualization tool, provides a dedicated 3D viewer for metabolic reconstructions, facilitating the visualization and comparison of complex metabolic maps.

Microbiome research aims to understand how microorganisms interact with their environments. Advances in metataxonomy and metagenomics now allow precise characterization of microbial communities, revealing their evolution and function. Metagenomics, in particular, captures the spatial, temporal, and functional complexity of microbes in situ, offering insights into gene expression and system-level functionality. Both approaches are applicable across diverse environments, from oceans to human samples, with adaptable DNA extraction protocols. For human studies, stool samples are common, while environmental samples use specialized filtration and preservation.

In this example, we apply the pilot CIESM 3D visualization tool, through its MetaCube prototype, to understand how different dietary patterns—specifically Western and Korean diets—affect the structure of microbial metabolic networks within gut microbiome samples. The Western diet is typically characterized by a high intake of animal proteins, saturated fats, refined sugars, and processed foods. In contrast, the Korean diet emphasizes plant-based foods, including vegetables, legumes, and fermented products, alongside rice, with a lower intake of animal fats and processed sugars.

These dietary differences were reflected in the metabolic networks reconstructed using MetaDAG, a web-based tool that compares metabolic networks from microbiome data (Palmer *et al.* 2025; available at <https://bioinfo.uib.es/metadag/>). As a demonstration, MetaDAG was applied to analyze 24 gut microbiome samples from Study MGYS00000394 in the MGnify database (Richardson *et al.* 2023). These samples originated from 12 individuals, each undergoing both Korean and Western dietary interventions, resulting in paired samples for comparative metabolic analysis. MetaDAG revealed distinct metabolic signatures associated with each diet, highlighting how dietary composition influences microbial community function. For instance, the Korean diet was linked to metabolic pathways related to fiber fermentation and short-chain fatty acid production, beneficial for gut health. Conversely, the Western diet showed enrichment in pathways tied to lipid metabolism and amino acid degradation, potentially associated with pro-inflammatory states.

The output datasets generated by MetaDAG, which MetaCube accepts, include core metabolic pathways (shared between diet types), pan-metabolic pathways (encompassing all samples), and individual sample datasets. The complexity of these metabolic maps makes them challenging to visualize and compare effectively in traditional formats. This is precisely where the pilot CIESM 3D visualization tool, realized as the MetaCube prototype, proves invaluable.

MetaCube serves as a dedicated 3D viewer for metabolic reconstructions, accepting reaction data obtained from MetaDAG. It functions as a dynamic reference map where each layer corresponds to a specific metabolic pathway, connecting enzymes involved in different pathways with straight vertical lines. Its fully navigable interface offers easy-to-learn, highly interactive 3D kinetics, leveraging the Three.js Javascript library and other open-source utilities under the M.I.T license.

As shown in Figure 6, we used MetaCube to visualize the core metabolism shared between Western and Korean diet samples. This CIESM-powered prototype allows us to clearly compare both core (left) and pan-metabolic (right) representations. For example, elements of pathway 00100 (steroid biosynthesis) are not commonly shared between the two datasets, indicating they are an accessory gene set.

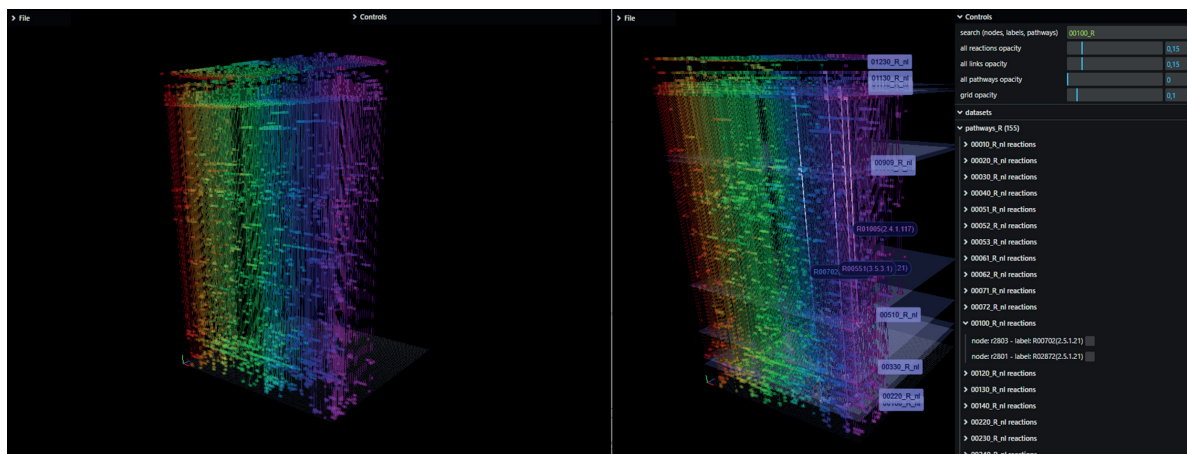


Figure 6. Core versus Pan metagenome representation.

The prototype MetaCube viewer also allows to identify differential pathways or gene features between two datasets. In the next depiction (Figure 7), we are considering the differences between the pan-metabolism of the Western (left) and Korean (right) diets.

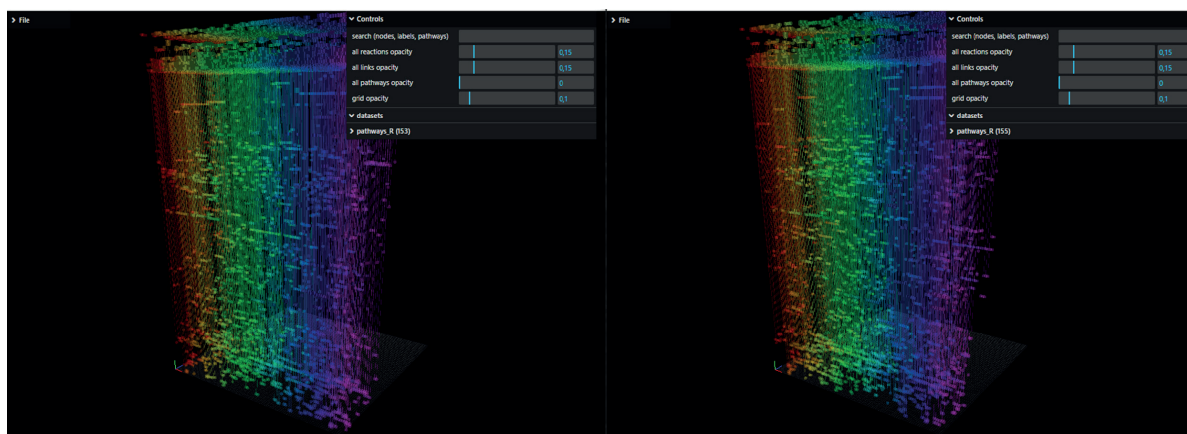


Figure 7. Pan-metabolisms from Western (left hand) and Korean (right hand) diets.

Even in its initial stages, the MetaCube interface, allows users to explore the intricate details of both datasets. This first version already enables highlighting differences in pathways or gene content by searching for specific features. The MetaCube viewer is a powerful visualization tool for metagenomics datasets, allowing us to identify common metabolic pathways or enzymes between samples. This tool, coupled with the reductive and comparative capabilities of MetaDAG, represents a comprehensive approach to metagenomics summarization and visualization.

Historically, the complexity of metagenomics results has made it difficult to establish general exploratory protocols. MetaCube, for the first time, offers a robust framework, to compare metagenomics datasets using a flexible reference platform. This platform allows for mapping every metagenomic result, either through a core or pan-metagenomic context.

MetaCube is currently hosted at metacube.ciesm.org.

6. VALUE OF OCEAN DATA

6.1. Legal implications and value chain of stakeholders

With the advancement of new technologies in visualization, modelling, and prediction, our society is entering a transformative era where ocean data play a critical role in the planning and safety of maritime activities, as well as in the design and implementation of environmental protection measures. In the face of current global crises—biodiversity loss, climate change, and pollution—access to high-quality ocean data and information is more vital than ever. These data are essential to advancing scientific understanding, particularly for complex marine phenomena such as ocean gradients.

However, significant barriers remain within the governance framework of ocean data collection, sharing, and use, which continue to impede scientific progress. Three main challenges can be identified, especially in relation to the use of cutting-edge technological tools such as 3D mapping, visualization, modelling, and predictive analysis for complex ocean phenomenon:

1. The nature of data collection activities
2. The modalities through which data are shared and used
3. The governance framework overseeing these processes

Despite substantial efforts made by States to enhance cooperation in sharing marine scientific data and information, the international legal framework still contains many ambiguities and loopholes. These ambiguities and loopholes complicate access to, and the collection, sharing and use of, ocean data & information. The marine scientific research (MSR) regime under the United Nations Convention on the Law of the Sea (UNCLOS) provides a foundational legal framework for conducting marine research and sharing related data & information. However, the difficulties encountered during the negotiation of UNCLOS left a key gap: the absence of a clear definition of “marine scientific research”. This gap has led to divergent interpretations among Coastal States regarding activities conducted within their national jurisdictions. As a result, many activities involving the collection and sharing of marine data face legal and administrative challenges, due to the divergent interpretation of States of what constitutes MSR.

Among the diverse range of ocean data activities, operational oceanography stands out as a particularly illustrative case. It has been the subject of extensive discussions within the Advisory Body of Experts on the Law of the Sea (ABE-LOS) of the Intergovernmental Oceanographic Commission (IOC) of UNESCO. While ongoing disagreements persist among States on the nature of operational oceanography, these discussions ultimately led to the development of an ad hoc legal framework governing the operation of Argo floats. Other emerging technologies used within national jurisdiction—such as satellite-derived data, ocean observations from submarine cables (*e.g.*, SMART systems), as well as drones or autonomous unmanned vehicles—are also currently facing important legal uncertainties both within and outside the European Union. These uncertainties often translate into challenges in obtaining operational authorizations or in collecting, sharing and using the data collected by such technologies. Coastal States, seeking to protect their sovereignty and sovereign rights (including knowledge of resources, activities, and geographic features, including for defence and security purposes), have imposed various limitations. These may include restrictions on data resolution, geographic coverage, or the sharing and final use of the collected data, as well as the ban of certain activities. As a result, a wide range of valuable ocean data which could be collected by advanced technologies and

infrastructures remains outside the scope of the freedom of marine scientific research.

In addition to issues related to data collection, broader challenges also arise concerning the modalities for sharing and using scientific data, particularly when it involves advanced technological tools that could incorporate artificial intelligence. These tools often rely on aggregated data from multiple sources and stakeholders (including private) for visualization, modelling and predictive analysis. A notable example is the Digital Twin of the Ocean, developed under the leadership of the European Union. These technologies introduce a complex set of questions, including:

- **Data ownership:** Who owns the derivative products generated by the raw data? How to navigate the initial ownership or co-ownership of the raw data (from the Coastal State and/or research operator depending on the collection point) and the new potential ownership of derivative products?
- **Model ownership:** Who holds rights over AI-generated models or digital twins?
- **Liability and responsibility:** Who is accountable for data accuracy, errors in prediction, or unintended outcomes - especially when prediction is designed for policy decision-making?
- **Traceability and auditability:** How can data usage and decision-making processes be transparently tracked across the data chain of stakeholders?
- **Jurisdictional and Regulatory Applicability:** What legal issues are triggered by the use of the Digital Twin for Ocean by non-EU Members? How can the non-applicability of EU regulations (such as the EU AI Act or Open Data Directive) be navigated, requiring data agreements to be negotiated with non-EU Member States (including CIESM Member States)?

Moreover, the dual-use of ocean data—where scientific information may also serve commercial, security, or strategic purposes—and the increasing value of data when layered or integrated (*e.g.*, through AI or multi-source modelling) are creating new international legal and ethical dilemmas. These issues underscore the urgent need for States to refine existing governance mechanisms to prevent further conflicts over data collection and use, especially in contexts of transboundary study and cooperation.

At the governance level, the ocean data landscape is becoming increasingly complex due to the growing number of public and private stakeholders involved across the data value chain. These include States (coastal and flag States), their national research institutes various sectors, including defense, private sector, data repositories (public and private), operators developing modelling and predictive systems (public and private), and end-users such as policymakers or the private sector. As this ecosystem expands, governing the collection, sharing, and usage of ocean data becomes more challenging.

Since the legal ambiguities and loopholes within the United Nations Convention on the Law of the Sea (UNCLOS) and related international law are unlikely to be resolved in the short to medium term, Member States of the Mediterranean Science Commission (CIESM)—gathering States Members and not Members of the European Union¹—could explore practical options

¹ This distinction is of importance as different legal and regulatory frameworks are applicable to these Member States of the CIESM. Notably, the recent AI Act of the European Union (first ever AI legal framework) is not applicable to non-EU Members, posing potential issues of various standards and protections regarding data quality, bias mitigation and liability, especially in the context of transboundary data sharing and multi-stakeholders systems.

to address current challenges and promote cooperative frameworks for data collection and exchange to improve knowledge of ocean gradients.

Several avenues could be considered, requiring further in-depth study. One pragmatic option would be to leverage the CIESM forum—a direct subject of international law and mandated to foster cooperation, promote scientific research, and facilitate data exchange. Within this context, CIESM could serve as a platform for its Member States to engage in dialogue on ocean data governance issues, with the aim of developing a CIESM Ocean Data Governance Policy. Such a framework could help harmonize approaches among Member States while leveraging the input of the Scientific Council and scientific experts.

In the longer term, Member States might also consider negotiating a specific agreement to address the current legal uncertainties in ocean data governance related to the mandate and scope of work of the CIESM. Such an agreement, which could be limited in scope, could support and strengthen scientific cooperation, facilitate data exchange and use, and promote mutual understanding across Member of the CIESM.

By adopting either approach, States would not only fulfil their legal obligations under international law—including the duties to cooperate, to exchange scientific data and information, and to protect and preserve the marine environment—but also contribute to a forward-looking and innovative model of regional ocean governance; adapted to the diversity of profiles of CIESM Member States. This would demonstrate the Mediterranean region’s commitment to modern, science-based cooperation in marine affairs, reinforcing CIESM’s role as a key facilitator in this evolving landscape.

6.2. Data Archives and Bioinformatics Resources

The generation of data from biological samples is often the result of costly experiments and sampling campaigns. In the case of marine data derived from either long-term and/or large-scale ocean sampling missions that require substantial financial investment and extensive planning, this is particularly true. Maximizing the value extracted from these samples and their corresponding datasets, which cannot easily be recaptured, is essential to justify the investment.

A key strategy for achieving this maximised value is to ensure that data are made openly available, following any appropriate embargo period to allow the primary researchers to publish their findings. Data should be deposited in public archives accompanied by comprehensive metadata about the sample and experimental protocols that provides essential context. For nucleotide sequencing data, the International Nucleotide Sequence Database Collaboration (INSDC) serves as the central repository, supporting data sharing across its member organizations, National Library of Medicine, National Center for Biotechnology Information (NLM-NCBI) (Robinson *et al.* 2024), European Molecular Biology Laboratory-European Bioinformatics Institute (EMBL-EBI) (Liang *et al.* 2023), and Research Organization of Information and Systems, National Institute of Genetics (ROIS-NIG) (Quinton *et al.* 2024). These recognized repositories offer long-term data preservation and promote best practices in data standards, metadata capture, data linking, and provenance, thereby facilitating future data reuse and integration into meta-analyses.

While sequence data repositories are well established and widely known, many other resources exist to archive diverse data types. For example, PRoteomics IDentifications Database (PRIDE) (Gohl *et al.* 2024) for proteomics, MetaboLights (Jeffryes *et al.* 2024) for metabolomics,

BioImage Archive and Electron Microscopy Public Image Archive (EMPIAR) for imaging (Gohl *et al.* 2024; Robinson *et al.* 2024), and Marine Environment Data and Information Network (MEDIN) data archive centre (DAC) (Marsh *et al.* 2024) for geology, geophysics and backscatter. Researchers are encouraged to deposit data in well-maintained public archives whenever possible to ensure data longevity and, critically, to enable linkage with associated datasets. The correct association of multi-modal datasets across resources is fundamental for scalable multi-omic analysis, and provides a basis for ocean modelling. However, as these various resources are independent of each other and therefore unaware of other associated omics datasets, this linking is most often achieved through the use of persistent sample identifiers. Resources such as BioSamples (Harris *et al.* 2024) provide stable identifiers that can be associated with data in other repositories, enabling the connection of diverse analyses performed on a single sample. As the BioSample record also contains sample metadata, such as geolocation and sampling data and time, it supports the linking of datasets in, for example, longitudinal studies.

In addition to describing raw data and samples, it is equally important to document any bioinformatic processing or analysis applied to a dataset. For example, metagenomic sequence data deposited in INSDC can be linked with taxonomic and functional profiles generated through specific bioinformatic pipelines. The interpretability of these profiles relies on transparent documentation of the analysis pipeline, including versioning and accessibility. Resources such as MGnify (Finn *et al.* 2024), Qiita (Zaneveld *et al.* 2018), and Integrated Microbial Genomes & Microbiomes (IMG/M) (Wilke *et al.* 2023) provide versioned, openly documented pipelines, ensuring full provenance for downstream users.

A wealth of marine data is available in public archives and is suitable for the types of studies and meta-analyses discussed during the workshop. However, heterogeneity in the associated metadata can limit data utility. Despite the existence of standards and guidelines—such as those from the Genomic Standards Consortium (GSC) (Filed *et al.* 2011; Gouldsbury *et al.* 2024) and the FAIR (Findable, Accessible, Interoperable, Reusable) data principles (Wilkinson *et al.* 2016), gaps in contextual metadata remain. It is essential to adopt and promote these best practices, capturing rich sample metadata to enhance dataset utility for future research. There are numerous large-scale initiatives underway, including DTO-BioFlow (GA:101112823), Blue-Cloud2026 (GA:101094227), BIOcean5D (GA:101059915) and BlueRemediomics (GA:101082304) to either generate or aggregate standardized, well-described marine data. These efforts offer substantial, high-quality datasets that can be expanded and leveraged to contribute to the aims of this workshop.

7. CONCLUDING NOTES

The workshop “An Ocean of Gradients - Towards a 3D Mapping and Visualization of High-Resolution Marine Data” highlighted the critical need to understand marine environmental heterogeneity and the complex dynamics of ocean ecosystems. Defining gradients as changes across various distances (spatial, temporal, and spatiotemporal) is fundamental to this understanding. Analyzing the growing volume of high-resolution marine data requires sophisticated visualization and quantification tools. The pilot CIESM 3D visualization tool marks a significant advancement, offering a versatile and user-friendly platform for visualizing and interpreting diverse marine data, including biological, physical, chemical, and even genetic parameters. By overlaying datasets onto 3D bathymetric maps, this tool enables volumetric

visualization of nutrient hotspots and enhances the detection of biogeochemical gradients, which is crucial for understanding the intricate interactions and environmental changes occurring throughout the Mediterranean Sea.

The pilot CIESM 3D visualization tool holds immense future potential. Its ability to easily incorporate and visualize data from extensive public repositories like GEBCO, EMODNET, MGnify, TARA, and ODATIS will lead to a more comprehensive and integrated approach to marine data mapping. This interoperability facilitates advanced visual exploration and analysis of generated niche profiles and associated data. Furthermore, the tool's capacity to develop detailed censuses of marine biological niches and contribute to a Microbiome MetaCube (MMC) promises to unify geological, physicochemical, and biological data into a globally accessible format. Crucially, the pilot CIESM 3D visualization tool is being developed with an emphasis on free access, promoting open science principles and maximizing the value of costly experimental and sampling campaigns.

Beyond the tool itself, the workshop also stressed the importance of improving ocean data governance. Significant barriers to data collection, sharing, and use, including legal ambiguities within the UNCLOS framework, continue to impede scientific progress. The dual-use nature of ocean data (for both scientific and commercial/security purposes) and the increasing value of integrated data also create new legal and ethical dilemmas. Addressing these challenges through cooperative frameworks, potentially leveraging CIESM as a platform for dialogue and policy development, is essential for informed conservation and management efforts. By ensuring open access and promoting best practices in data archiving and metadata capture, the scientific community can enhance data utility for future research and contribute to a forward-looking model of regional ocean governance.